HDOC Day-52

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BATCH-15

QUESTION:

Write a C program that reads the size and elements of a variable-length integer array arr[]. Pass the array to a function named previous_greater() using its address.

Using pointers and nested loops, find the nearest previous greater element for each element of the array. Search only the elements to the left of the current element.

If no greater element exists to the left, display -1. Display all results in their order of appearance, separated by commas.

Aim:

To write a C program to find the previous greater element of each element in an array using pointers, functions, and nested loops.

Algorithm:

Step 1: Start the program.

Step 2: Read the size n of the array.

Step 3: Declare an integer array arr[n].

Step 4: Read n elements into the array.

Step 5: Pass the array to the function previous_greater() using its address.

Step 6: Use an outer loop to traverse each array element.

Step 7: Initialize ans = -1 for each element.

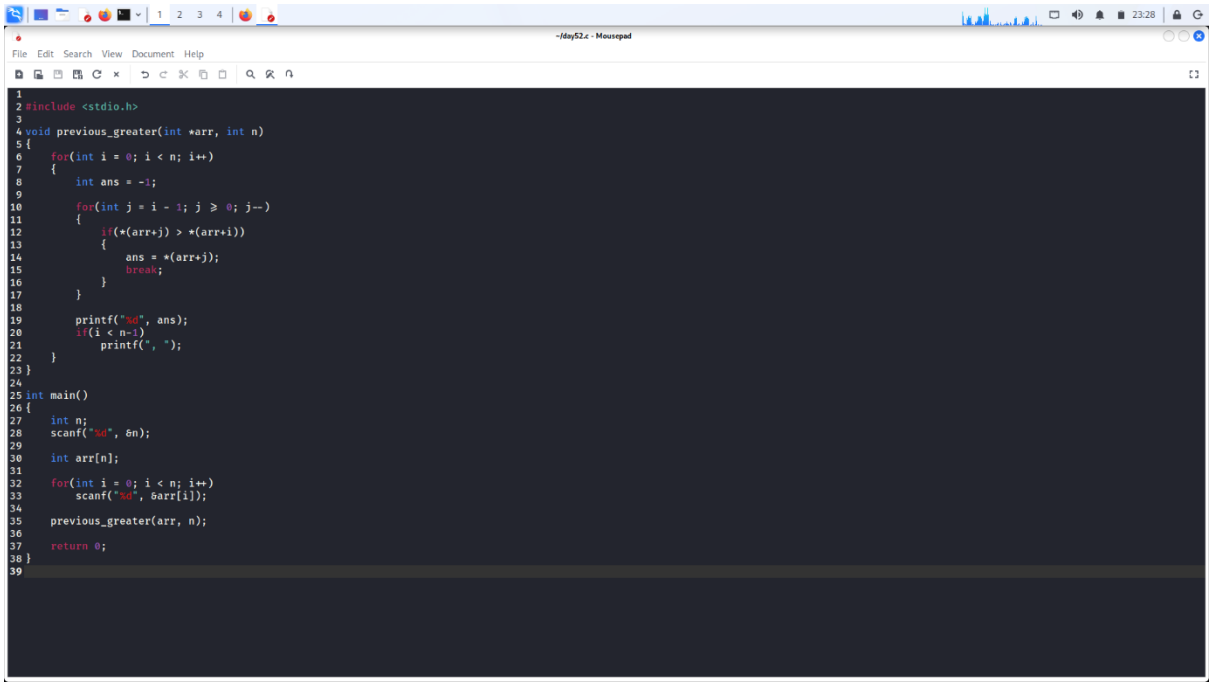
Step 8: Use an inner loop to search the elements on the left.

Step 9: If a greater element is found, store it in ans and break the loop.

Step 10: Print ans for each element, separated by commas.

Step 11: Stop the program.

>C PROGRAM;



```c
#include <stdio.h>

void previous_greater(int *arr, int n)
{
    for(int i = 0; i < n; i++)
    {
        int ans = -1;

        for(int j = i - 1; j >= 0; j--)
        {
            if(*(arr+j) > *(arr+i))
            {
                ans = *(arr+j);
                break;
            }
        }

        printf("%d", ans);
        if(i < n-1)
            printf(", ");
    }
}

int main()
{
    int n;
    scanf("%d", &n);

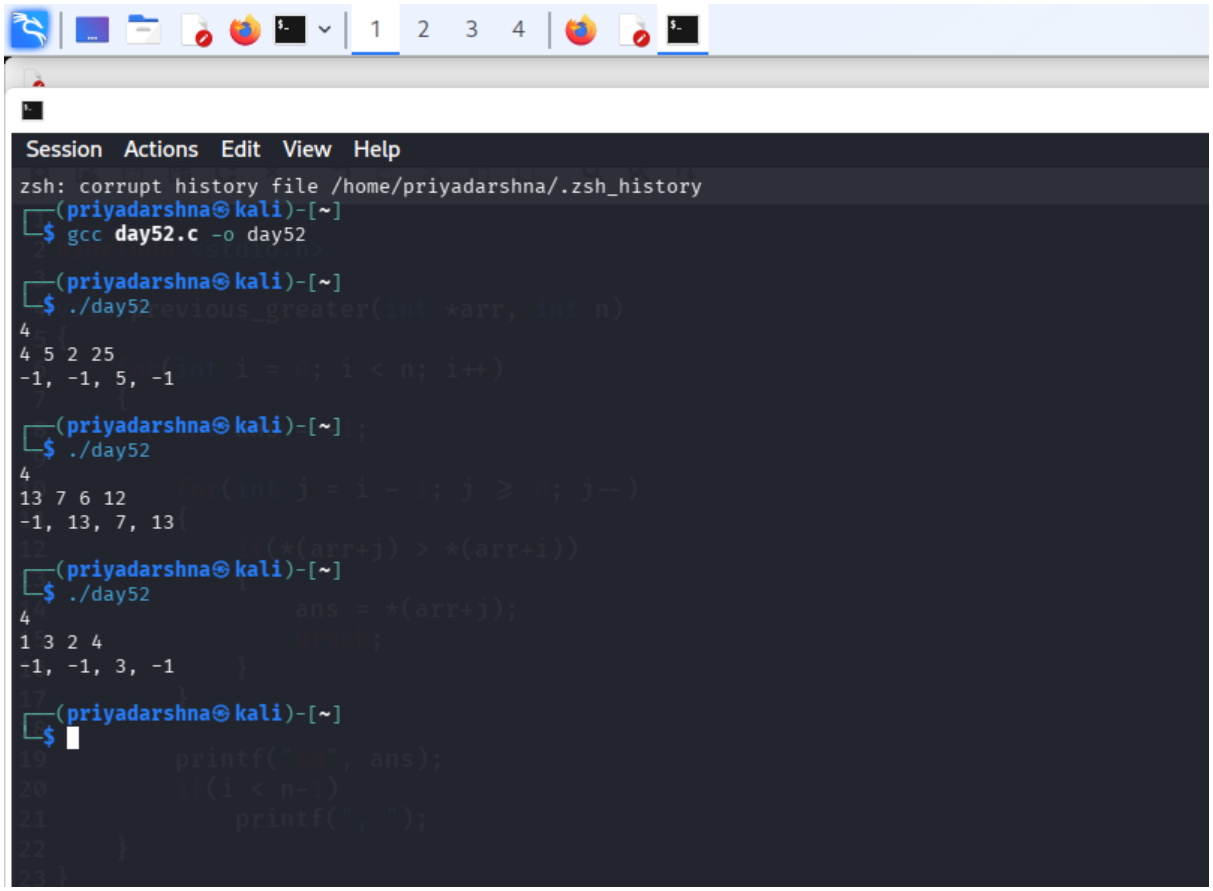
    int arr[n];

    for(int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    previous_greater(arr, n);

    return 0;
}
```

>OUTPUT;



```
Session  Actions  Edit  View  Help
zsh: corrupt history file /home/priyadarshna/.zsh_history
┌──(priyadarshna㉿kali)-[~]
└─$ gcc day52.c -o day52

┌──(priyadarshna㉿kali)-[~]
└─$ ./day52
4
4 5 2 25
-1, -1, 5, -1

┌──(priyadarshna㉿kali)-[~]
└─$ ./day52
4
13 7 6 12
-1, 13, 7, 13

┌──(priyadarshna㉿kali)-[~]
└─$ ./day52
4
1 3 2 4
-1, -1, 3, -1

┌──(priyadarshna㉿kali)-[~]
└─$
```